Mersenne Scaling in Cyclic Rule 90

Route-A structural certificate C150

Abstract

For Rule 90 on every binary ring of Mersenne circumference $L = 2^r - 1$, the local multiplier $a = x + x^{-1}$ satisfies $a^{L+1} = a$. Its image contains exactly half the state space, every state reaches it in one step, and all image points are periodic with periods dividing L. Polynomial gcd and Möbius inversion resolve primitive cycles. In contrast, power-of-two rings are nilpotent. This is an exact scaling theorem, not a target determinant.

1 Mersenne identity and image

Work in $R_L = \mathbb{F}_2[x, x^{-1}]/(x^L - 1)$, where one Rule-90 update is multiplication by $a = x + x^{-1}$. If $L = 2^r - 1$, Frobenius gives

$$a^{2^r} = x^{2^r} + x^{-2^r} = x + x^{-1} = a, \qquad a^{L+1} = a. \tag{1}$$

Multiplication by x is invertible and $xa = (x+1)^2$. Since odd L makes $x^L + 1$ squarefree and $x + 1$ is a factor, $\gcd(x^L + 1, (x+1)^2) = x + 1$. Hence $\dim\ker a = 1$ and $\dim\operatorname{im} a = L - 1$.

For $y = au$ in the image, (1) yields $a^L y = y$. Thus the restriction to the image is a permutation with order dividing L. Every state enters the image after one update, and its 2^{L-1} points are periodic: exactly half of all states.

2 Exact periods

The standard multiplication-kernel calculation gives

$$\mathrm{Fix}_L(n) = 2^{\deg\gcd(x^L+1,(x^2+1)^n+x^n)}. \tag{2}$$

Consequently

$$P_L(n) = \sum_{d|n} \mu(n/d)\,\mathrm{Fix}_L(d), \qquad C_L(n) = P_L(n)/n.$$

All realized periods divide L.

3 Power-of-two control and boundary

For $L = 2^s$, $a^{2^{s-1}} = x^{2^{s-1}} + x^{-2^{s-1}} = 0$ because the two monomials coincide. Thus the map is nilpotent and only zero is periodic.

Exact ledgers through $r, s \le 8$ pass 153 independent matrix assertions, 276 SymPy checks, byte replay, and 45 hostile cases. The strict tuple is ($\texttt{A1\_WEAK}, \texttt{A2\_FAIL}, \texttt{A3\_FAIL}, \texttt{A4\_FAIL}$). We claim no target divisor or counting law, arithmetic/local factor, root number, automorphy, Hilbert–Pólya operator, or Route-B authorization. Scope: `NO_BAD_EULER_OR_ROOT_NUMBER`.